Yuanfan Chen

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Education

Cornell Tech, Cornell University — New York, NY
Master of Engineering in Computer Science — *Aug. 2025 – May. 2026*

University of Toronto — Toronto, Canada
Honours Bachelor of Science, Computer Science Specialist (GPA: 3.87/4.0) — *Sep. 2020 – May. 2024*

Personal Experience

Machine Learning System Researcher — Sep 2025 – Present
Cornell University — *New York, NY*

- Engineered a scheduling framework integrating **RCPSP modeling**, **EDF**, and **Greedy** to maximize throughput under **multi-class SLO constraints**; derived **stepwise performance functions** via GPU profiling and validated performance by reproducing SOTA baselines (**Slo-serve**, **Sarathi**).
- Leveraged **γ-Boost Queuing Theory** to analyze **light-tailed workloads**, optimizing resource scheduling for prefill versus decode phases; achieved a **300% tail latency improvement** while maintaining Avg E2E and TTFT parity with **SJF (Shortest Job First)**.
- Developed a **Vidur-based simulator** for **Multiclass SLO** evaluation, conducting a class-level comparative analysis of **Colocation vs. Separation** global scheduling strategies and reproducing **PolyServe** as a baseline.

Kernel Development Intern — Jul 2025 – Aug 2025
*Tencent: **TCHouse-X Distributed Database*** — *Beijing, CN*

- Designed and implemented an **admission control mechanism** to regulate request concurrency, ensuring system stability under bursty loads (which later inspired my Multi-SLO research).
- Contributed to the open-source **ClickHouse-X** project; established rigorous **CI/CD pipelines** and code review standards to build robust, maintainable, and reproducible distributed systems.
- Optimized node recovery via fault-tolerant **KV delta-apply**, reducing latency from **10s to 3s** and ensuring data consistency in high-churn clusters.

Machine Learning Engineer — May 2025 – Sep 2025
*Tencent: **Distributed Training and Inference*** — *Beijing, CN*

- Mastered parallelization strategies including **Tensor Parallelism (TP)** and **Data Parallelism (DP)**, and optimized inference engines using **PagedAttention** and **Continuous Batching**.
- Optimized **GQA** and **DSA** kernels on **NVIDIA H200 and B200** GPUs to accelerate LLM inference; profiled memory hierarchies and warp stalls using **Nsight Compute (NCU)** to alleviate bandwidth bottlenecks and maximize hardware utilization.
- Reduced pipeline bubbles by implementing **Chunk Prefill** and batched decode strategies, significantly improving throughput for long-context inference workloads.

Optimized **MLA** kernels on **NVIDIA H200 and B200** GPUs to accelerate LLM inference; profiled memory hierarchies and warp stalls using **Nsight Compute (NCU)** to alleviate bandwidth bottlenecks and maximize hardware utilization.

Reduced pipeline bubbles by implementing **Chunk Prefill** and batched decode strategies, significantly improving throughput for long-context inference workloads.

Database Researcher — Jan 2024 – Feb 2025
Far Data Lab — *Toronto, CA*

- Co-designed **dpBento** to benchmark compute/memory/network/storage of heterogeneous DPUs for data processing offload.
- Studied **RocksDB** internals; explored DPU offload for decompression/compaction to evaluate compute–storage disaggregation.

Technical Skills

Programming: Python, C++, CUDA, Java **AI Infra**: vLLM, SGLang, KV-Cache, PD separation
Distributed: CAP, ClickHouse, HDFS **Databases**: RocksDB, MySQL, Redis
Networking: epoll, multiplexing, async I/O, KCP **Systems**: Linux internals, CPU/GPU arch, Nsys/Nsight

Publications

- **dpBento: Benchmarking DPUs for Data Processing**. Jiasheng Hu, Chihan Cui, Anna Li, Raahil Vora, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. arXiv:2504.05536 (2025).

Academic Service

- **Invited Reviewer**: IEEE Journal on Selected Areas in Information Theory (JSAIT) – Special Issue on Energy and Data Efficiency in AI (2026). Reviewed submissions focusing on cutting-edge LLM inference systems.